

Lesson code		2022-08-01	Level	Elementary	Total Time	40mins	Teaching Week	Aug (01-05)	
Subject		Science	Books	Science Fact File 3(2 nd Edition)		Page No.	02-08	Class	VIII
Day Time	Chapter Topic	Objectives: The students will be able to:	Teaching Resources:	Teaching Learning Strategies					
Day 1 40min	Unit 1 Work and energy Work	Interpret about work and its units	Textbook, notebook	Introduction - On the 1st day of the class, interact with the students in a friendly manner. - Introduce yourself and about the subject. - Ask the names of the students. Methodology (Activity based learning) - Ask the students to define work, listen to their answers and write them on the board. - Write the definition of work on the board. - Ask any student to come in front of the class and push the table, now ask the students did he/she use any force to do this work. - Close the door of the class room and ask another student to come in front and try to open the door. Which force he/she uses for opening? - Explain that the amount of work depends on the force and distance. - Discuss the units of work. - Ask the students to answer the questions given on textbook page 03 and note down in their notebook. Wrap up Recap of the lesson by asking questions; - What is the unit of work? Home Work: Revise the topic, go through the written work and learn the answers done the in class.					
Day 2 40min	Unit 1 Work and energy Energy and work	Comprehend the relation of work and energy	Textbook, worksheet	Introduction To recall the prior knowledge, pick any of two students randomly and ask them to write the answers of following questions on board. - Define work - How the work can be measured? Methodology (Discussion) - Ask the students if a car has no fuel in it, can you drive that car? - Explain that our bodies need food to work like the car needs fuel to drive. - Explain the units of energy and work. - Tell the students that there are several different types of energy which can be used to do work. - These can be conveniently divided into two groups- potential energy and kinetic energy. - Let the students solve the worksheet. Wrap up Recap of the lesson by asking questions; - Is the unit of work and energy are same?					

				• 1 MJ =? , 1 kJ =? Home Work: Give one of the worksheet as a homework.
Day 3 40min	Unit 1 Work and energy Potential energy	Define potential energy and brief gravitational potential energy	Textbooks, balls of different sizes, stop watch	Introduction To recall the prior knowledge, ask the following questions from the students: • What units are used for measuring work and energy? Methodology (Activity based learning and discussion) • Write the definition of potential energy and the names of its types on board • Explain the gravitational potential energy with respect to the story of Sir Isaac Newton about gravity. • Make a queue and take the students to the playground • Make groups of four or five and give them two balls of different sizes • Ask the student to throw the ball with less weight in the air, the other student will note the time on the stop watch • Vice versa for the ball with more weight • Ask the students to compare the timings of the fall of the two balls. • Explain that the weight of an object is the force of gravity pulling down on it. The greater the force of gravity pulling on the object, the greater its weight. • Discuss the answers of questions on textbook pg. 05 • Students will note down their answers in notebooks as a homework. Wrap up Recap of the lesson by asking questions; • How did Sir Isaac Newton discover gravity? Home Work: Note down the answers of questions in notebooks as discussed in the class.
Day 4 40min	Unit 1 Work and energy	Explain elastic, chemical and nuclear potential energy.	Textbooks, elastic bands, coil springs	Introduction To recall the prior knowledge, following questions will be asked from students: • Name different types of potential energy. • What is meant by gravitational potential energy? Methodology (Activity based learning) • Make groups of students and give them elastic bands and rubber bands • Ask the students to stretch the elastic bands • Explain that elastic potential energy is the energy stored in an object when deformed by being pulled or pushed • Vice versa with coiled springs • Write the definitions of chemical energy and nuclear energy on the board • Explain the difference between fission reaction and fusion reaction. • Students will give the examples of fission and fusion reaction and teacher will write it on the board, • Discuss the questions mention on textbook page 07. Wrap up Recap of the lesson by asking questions; • Differentiate between nuclear fission and fusion reaction.

				Home Work: Note down the answers of the questions mentioned on textbook page 07 as discussed in the class.
Day 5 40min	Unit 1 Work and energy	Interpret about kinetic energy	Textbook	Introduction To check the prior knowledge, following activity should be performed. - Students will be encouraged to present short presentations on the types of potential energy. Methodology (Discussion) - Define kinetic energy. - Tell the students that the word “<i>kinetic</i>” comes from Greek work ‘kinesis’ which means ‘moving’. - Explain with examples that kinetic energy depends on mass and velocity - Explain that by doubling the speed of a moving object, its kinetic energy is increased four times. - Discuss the answers of question on textbook page 08 in groups. - Student will write the answers in their notebook. - Let the students solve the worksheet. Wrap up Recap of the lesson by asking questions; - Name the two things on which kinetic energy depends upon? Home Work: Revise the topic and complete the written work.



Allied Schools Worksheet

Name: _____

Date: _____

Class: VIII

**Ch 1: Work and
energy**

Q1. Write the definition of 'work'.

Q2. What is the unit of work?

Q3. When is one unit of work done?

Q4. A donkey pulls a cart with a force of 100 N over 2 km. How much work has the donkey done?

- Q5. A car with two men in it broke down. For 20 minutes they tried to push it to the nearest garage which was 500 m away. One man pushed with a force of 200 N, the other with a force of 250 N. However, the car was too heavy and did not move. How much work did they do?**

ANSWERS

A1: Work is done when a force is used to make something move.

A2: Joule or kilo joule

A3: 1 joule of work is done when a mass of 1 N is moved over a distance of 1 m.

A4: $\text{work} = \text{force} \times \text{distance}$

$$\begin{aligned}\text{donkey's work} &= 100 \text{ N} \times 2000 \text{ m} \\ &= 20,000 \text{ joules} = 200 \text{ kJ}\end{aligned}$$

A5: No work was done since the car did not move. (It is likely that the men are tired though!)



Allied Schools Worksheet

Name: _____

Class: VIII

Date: _____

Ch 1: Work and
energy

- Q1. A boy pushes box 2 metres along a smooth surface. He then pushes the same box 2 meters along a rough surface. In which situation will more work be required? Why?

- Q2. Complete the following:

1 KJ = _____ J

1 MJ = _____ J

- Q3. Calculate the work done in the following situations. Take $1 \text{ kg} = 10 \text{ N}$

- i. to lift a 10 kg school bag from the ground to the top of the bus 3m high

- ii. to carry a 30 kg suitcase up to the 4th floor if each floor is 2m high

- iii. when a force of 5N pulls a book along a table for a distance of 40cm

iv. to carry a 2kg school bag from the classroom to the school gate 30m away

v. when a 20N fruit is falling freely from the top of a tree 300 cm high

vi. State the energy changes that take place in question v.

ANSWERS

A1: On rough surface more force is applied so more work will be done.

A2: Complete the following:

$$1 \text{ KJ} = \underline{1000 \text{ J}}$$

$$1 \text{ MJ} = \underline{1000,000 \text{ J}}$$

A3: Work done = force $\times$ distance

i. $10 \times 10 = 100 \text{ N}$ $100 \times 3 = 300 \text{ Nm}$

ii. $30 \times 10 = 300 \text{ N}$ $300 \times 4 \times 2 = 2400 \text{ Nm}$

iii. $5 \times 40 \text{ } 100 = 2 \text{ Nm}$

iv. $20 \times 300 \text{ } 100 = 60 \text{ Nm}$

v. Potential energy to kinetic energy